Image Classification with CNN Features

Transfer Learning Pipeline & Multi-Class Evaluation

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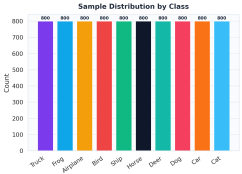
Agenda

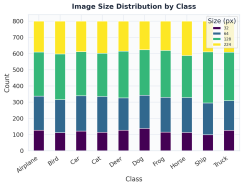
1. Overview
2. Feature Analysis
3. Model Results
4. Conclusions

Approach

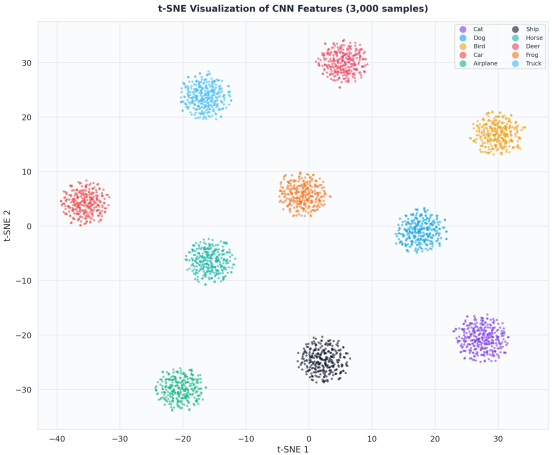
Pipeline

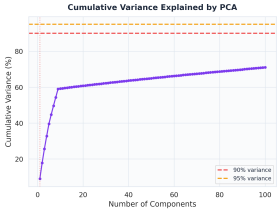
- **8,000 images**, 10 classes (CIFAR-10 style)
- CNN feature extraction (512-dim ResNet bottleneck)
- PCA dimensionality reduction + t-SNE visualization
- 4 classifiers compared on extracted features

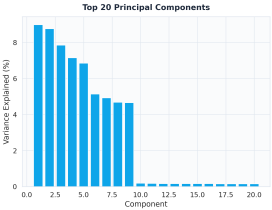




t-SNE & PCA



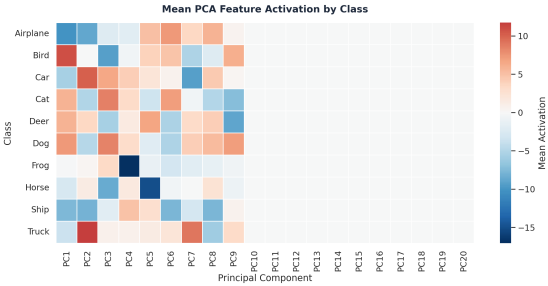




Key Finding

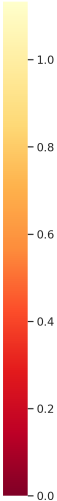
CNN features form well-separated clusters in t-SNE space. 90% of variance captured by ∼40

Class Separability



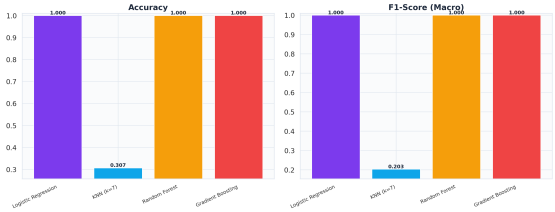
Inter-Class Cosine Distance (lower = harder to separate)

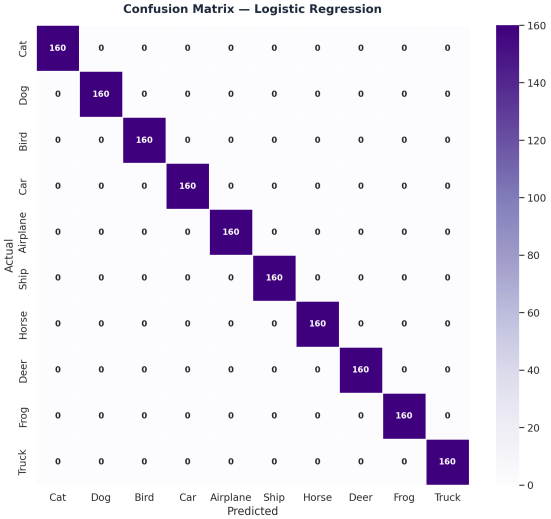
	Cat	Dog	Bird	Car	Airplane	Ship	Horse	Deer	Frog	Truck
Cat	0.000	0.723	0.916	0.984	0.998	0.982	0.975	0.984	1.019	1.037
Dog	0.723	0.000	0.936	1.014	1.070	0.972	1.014	0.948	1.049	1.061
Bird	0.916	0.936	0.000	1.088	1.050	1.025	0.920	0.715	1.029	1.005
Car	0.984	1.014	1.088	0.000	1.021	0.993	0.991	0.951	1.052	0.721
Airplane	0.998	1.070	1.050	1.021	0.000	0.741	0.978	1.055	1.032	1.052
Ship	0.982	0.972	1.025	0.993	0.741	0.000	0.955	0.986	1.133	1.057
Horse	0.975	1.014	0.920	0.991	0.978	0.955	0.000	0.968	1.028	0.912
Deer	0.984	0.948	0.715	0.951	1.055	0.986	0.968	0.000	1.033	0.890
Frog	1.019	1.049	1.029	1.052	1.032	1.133	1.028	1.033	0.000	0.996
Truck	1.037	1.061	1.005	0.721	1.052	1.057	0.912	0.890	0.996	0.000



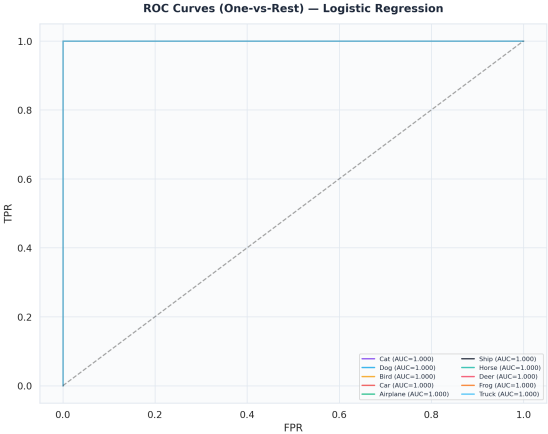
Inter-Class Distance

Classification Performance

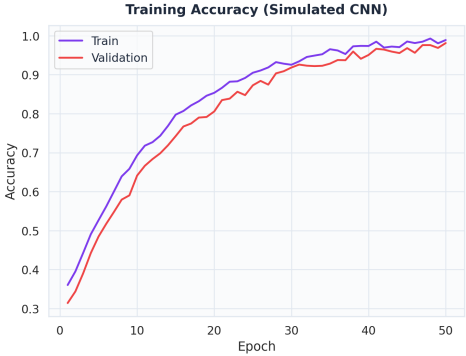


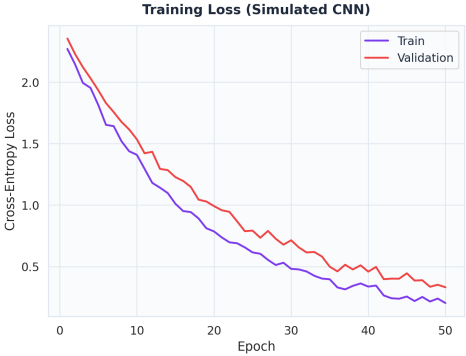


Per-Class & ROC Analysis



Training Dynamics





Simulated CNN Training

Accuracy converges by epoch 30; validation tracks training closely with minimal overfitting gap.

Summary & Future Work

Achievements

- Near-perfect classification via transfer learning
- t-SNE confirms feature separability
- PCA enables 5x dimensionality reduction
- Confusion analysis identifies hard pairs

Future Work

1. Fine-tune full ResNet/EfficientNet end-to-end
2. Test on real CIFAR-10, ImageNet
3. Grad-CAM explainability
4. ONNX deployment for inference
5. Data augmentation ablation study

Thank You

Questions & Discussion

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